Faster tests, greener cars

Can we predict how long a Mercedes-Benz car stays on the test bench?

Mohammad Bajbouj · ML project (Kaggle: Mercedes-Benz Greener Manufacturing) · neuefische AI PM bootcamp · October 2026

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Business case, approach and expected outcome

Main · BC

Business case

- Every Mercedes-Benz car is tested on a test bench before delivery
- Test times differ from car to car: 72 to 265 seconds
- So the test bench is hard to plan – waiting and overrun cost time and energy

Approach

- Public Mercedes-Benz data: 4,209 tested cars, 376 columns
- Test our hypothesis with identical cars: same configuration = same test time?
- Predict the test time with three machine-learning methods, keep the simplest

Expected outcome

- A prediction of each car's test time from its configuration
- A clear answer to our hypothesis: how much of the test time can be predicted at all
- A proposal for planning the test bench, with buffer and next steps

Source: business case: framing_h2_v3.md; data: D4, D17; H2: D13; three methods: D15, D33; slide added: D70

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Our hypothesis (H2)

Main · H2

H2: Does the same car configuration always need the same test time?

Why it matters

If identical cars get different test times, that part cannot be predicted from the configuration – it sets a limit for every model.

How we test it

Find cars with identical configurations (all 376 columns the same) and compare their test times. Then predict with three different algorithms: Ridge, Random Forest and XGBoost.

Where we answer it

Identical cars: slide 6. Three algorithms compared: slide 12.

Meaning: one hypothesis, three different algorithms – as the coaches asked.

→ *explained on slides 18–19* *Source: H2 chosen: D13; three algorithms: D15, D33; question: framing_h2_v3.md; slide added: change 11 (D59)*

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M0 – the data

Main · M0

4,209

tested cars

376

columns per car (8 text + 368 yes/no (0/1)); Mercedes-Benz hid the real names and replaced them with X0, X1, X2 …

72–265 s

test time per car (what we predict): shortest 72.11 s, longest 265.32 s

Meaning: we have the official Kaggle data – each car's configuration and its measured test time (→ slide 20).

→ explained on slide 20 *Source: content list #5 (D37); 4,209 cars: Step 10; 376 columns: Step 1 (D17); 72.11 s / 265.32 s: D50, Step 9 (D23); data: D4; merged with #4: change 14 (D59)*

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M1 – what the data tells us

Main · M1

1. Even identical cars get different test times → at most about 60 % of the differences in test time can be explained by the configuration (→ slide 23)
2. Most test times cluster around 75, 90, 100 or 108 seconds (→ slide 7)
3. One test of 265 seconds (next longest: 170 seconds) was removed as a likely faulty test (→ slide 7)
4. One column, X0, sorts the cars into these time groups – alone it explains 57.5 % of the differences (→ slide 8)
5. Of 368 yes/no (0/1) columns, 65 never change, copy or mirror another column; the strong ones only repeat what X0 already says (→ slide 28)

Meaning: at most about 60 % of the differences can be predicted, and the column X0 alone already explains 57.5 %.

→ explained on slides 26–28 *Source: content list #14 (D37; pointers: change 25, D64); 60 %: Steps 5–6 (R8); groups: Step 11 (R18); faulty test: Steps 8–9 (D23); X0: Steps 11–12 (D25); yes/no: Steps 13–16 (D27, D29)*

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Key finding: same car ≠ same time

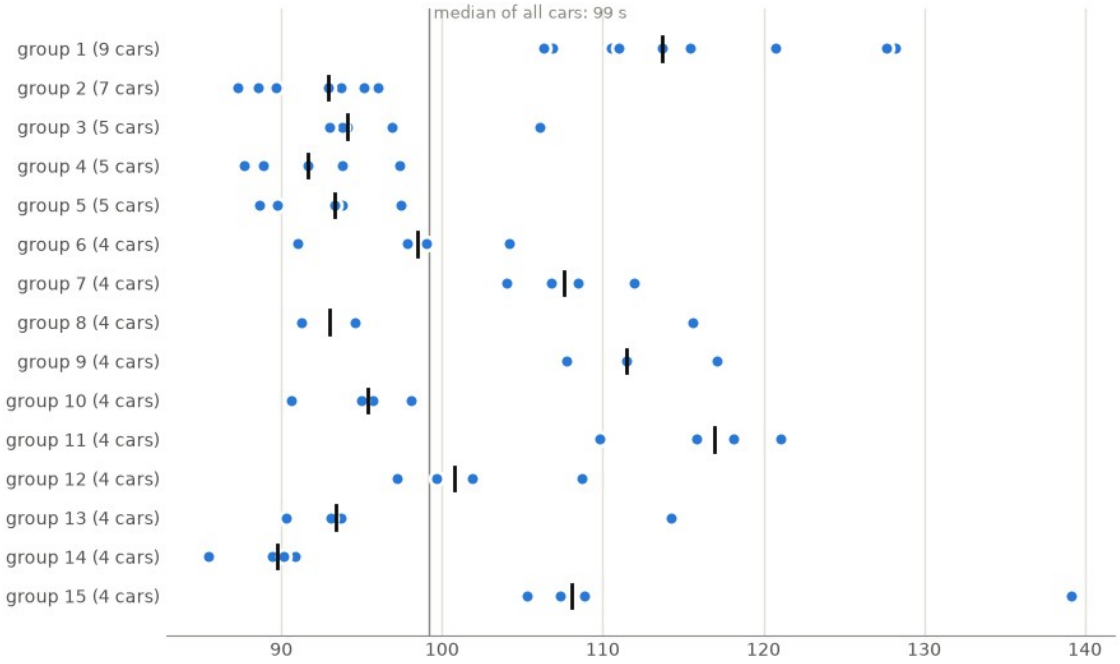
Main · M1

≈ 60 %

Using the configuration as indicator, we can predict up to about 60 % of the differences in test time between different cars (range 60–70 %; slides 23–24).

Each row = one group of identical cars; each dot = one car. Top row = the 9 identical cars (slide 22).

Expert view (Mohammad Bajbouj, automotive R&D): even identical cars can get different test times for reasons the data does not record, e.g. human error.

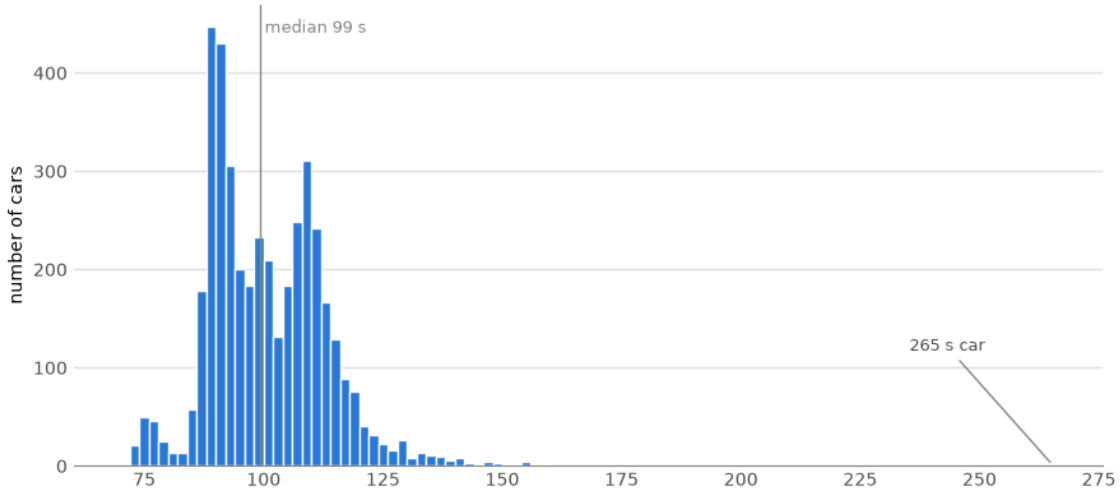


→ *explained on slides 22–24* *Source: content list #6; plot: Step 4 (R5); ≈ 0.6, range 0.6–0.7: Steps 5–6 (R6, R8, R9; D18, D19); expert view: his statement, change 13 (D59); 60 % sentence: change 38 (D67)*

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Test times come in groups, one unusual test

Main · M1



test time (seconds) of all 4,209 cars, 2-second groups

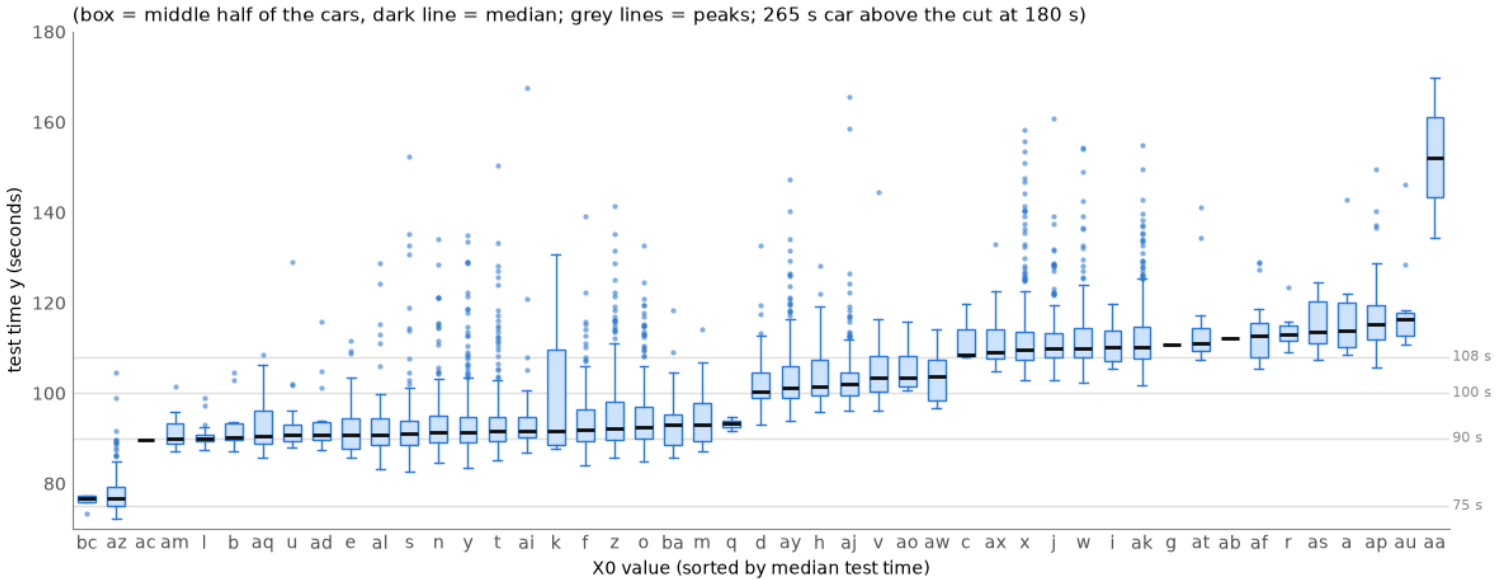
- Most cars need about 75, 90, 100 or 108 s
- One car took 265 s (next longest 170 s) → removed from training as a likely faulty test (our assumption – no column records it)
- The other 49 long tests stay

Source: content list #8; peaks: Step 11 (R18, D28); 265 s car ID 1770: Step 9 (R13, D23); 169.91 s: Step 5 (R7); 49 long tests: Step 8 (R11, D22); assumption: R12; histogram: Step 7 (R10)

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One column sorts the cars

Main · M1



Meaning: X0 alone explains 57.5 % of the differences (measured on the same cars it describes); the column X2 explains 22.6 %, every other text column at most 4.2 % (→ slide 25).

→ *explained on slide 25* *Source: content list #10; X0 57.5 %: Steps 11–12 (R20, D25); X2 / other text columns: key M1.5 result (D25); box plot: Step 11 (R19, y-axis cut at 180 s)*

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M2 – first predictions

1. Predicting the same average time for every car misses by about 12.4 seconds and explains nothing (0 %) (→ slide 10)
2. Predicting the average of the car's X0 group cuts the typical miss from 12.43 to 7.97 seconds – about 4.5 seconds better per car – and explains 58.7 % of the differences (→ slide 10)
3. The remaining 8 seconds are the noise floor: identical cars still differ by 8.1 seconds, so no model can explain much more than the limit of 59.7 % (→ slide 23)
4. A real model (Ridge) with X0 alone gives exactly the same result as the X0 group averages (7.97 seconds, 58.7 %) – the same rule, written as a formula (→ slide 10)
5. Adding all 310 columns (376 − 66 removed in M1) does not help: typical miss 7.97 → 8.09 seconds (→ slide 30)

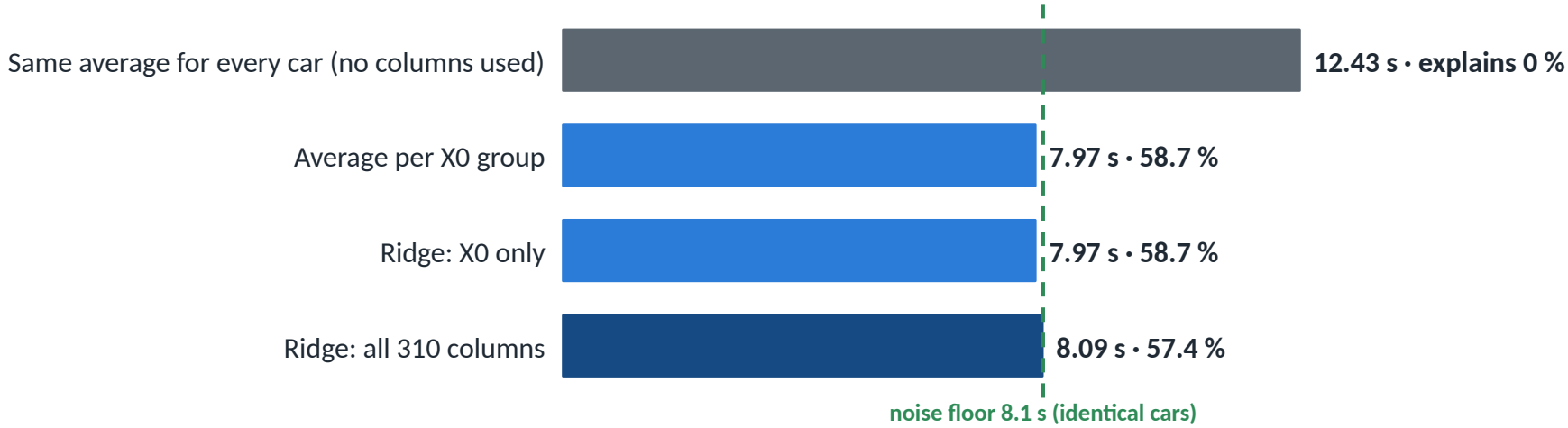
Meaning: X0 carries almost all predictable information; the rest is test-bench scatter the data does not record.

→ explained on slide 30 *Source: content list #16 (D37; reworded: changes 21, 22, D59; 24, 25, 28, D64); 4.5 s = 12.43 − 7.97 = 4.46 s: Steps 19–20; 8.1 s noise: Step 5, handover 28; 12 s / 8 s / 0.59: Step 19; limit: Steps 5–6 (R8, R37); Ridge 0.587 vs 0.574: Step 20 (R47); D32*

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Baseline: simple predictions

Main · M2



Typical miss in seconds · share of the differences explained. Our own test: 5-fold cross-validation on 4,208 cars (→ slide 33). 58.7 % is about 98 % of the limit 59.7 %. The official Kaggle test: slide 14.

Meaning: knowing only X0 cuts the typical miss by about 4.5 seconds per car (12.43 → 7.97 s); the remaining 8 seconds are the noise floor.

Source: content list #15; chart redrawn with plain labels from Step 21 values (changes 26–28, D64); Step 19 (R49); Step 20 (R47); chart: Step 21 (R50); 98 %: handover 16 result; CV / Kaggle: D44; D32

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M3 – best model

1. Ridge, Random Forest and XGBoost all miss by about 8 seconds (4.5 s better than one average for all) and explain 58.7 %, 58.9 %, 59.3 % – equal (→ slide 12)
2. So we keep the simplest model: Ridge, using only the column X0 (→ slide 32)
3. Tested on cars the model had not seen: for 62 of every 100 cars the prediction is at most 5 seconds off, for 91 of every 100 at most 10 seconds (→ slide 13)
4. The 10 biggest misses are all long tests (over 136 seconds) that took longer than predicted – they cannot be predicted from the configuration (→ slide 35)
5. Official Kaggle test on 4,209 hidden cars: our model explains 54.1 %, the winner 55.6 % → 97 % of the winner and 91 % of the limit, with one column (→ slide 14)

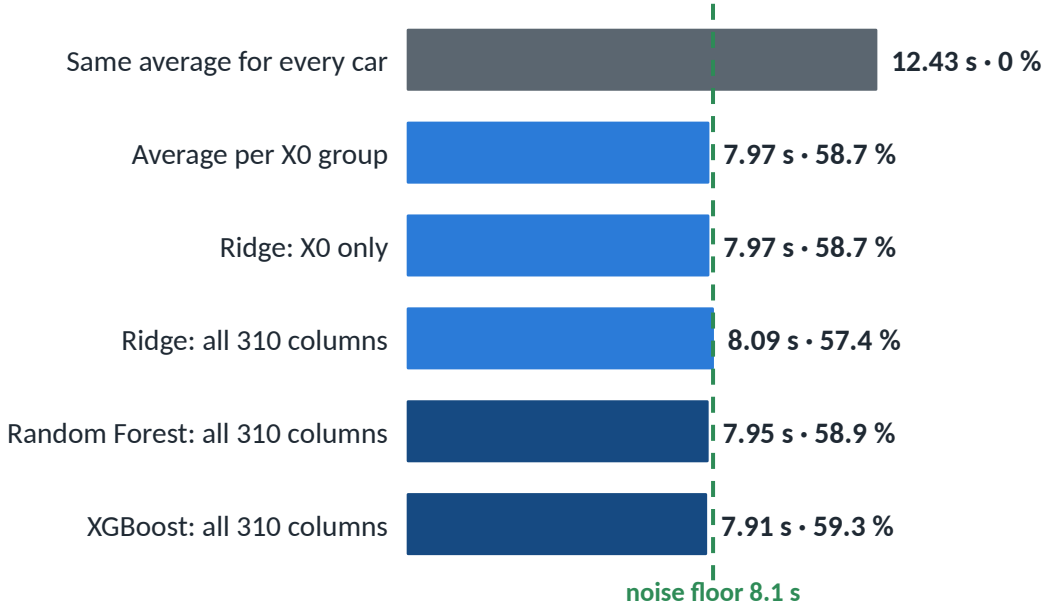
Meaning: the data, not the choice of algorithm, sets the limit – this answers H2.

Source: content list #21 (D37; reworded: changes 22, 23, D59; 25, 27, 28, D64); long tests not predictable: Step 29 (D61); test basis ±5 / ±10 s: Step 26 (out-of-fold, 4,208 cars); Steps 20, 23, 24 (D40); final model D43; ±5 / ±10 s: Steps 26–27 (D42); Kaggle 0.54107 vs 0.55550: D47, D44; 91 % = 0.54107 / 0.5965: handover 23 Part B (D50)

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Three models compared

Main · M3



Typical miss · share explained (our own test, 4,208 cars)
All three miss by about 8 seconds. The differences (at most 0.6 percentage points) are smaller than the normal variation between the 5 test rounds (about ± 3.5 points) → equal. (→ slide 33)

Meaning: we keep the simplest model: Ridge with X0 only.

→ *explained on slides 32–34* *Source: content list #19; chart redrawn with plain labels from the Step 25 table (changes 26, 27, D64); R^2: Steps 20, 23, 24 (D40); ± 0.035–0.042 per model: handover 28 (D58); seconds per model: Step 25 chart; equal within noise: R54, D40; final model: D43*

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Where the model fails

Main · M3

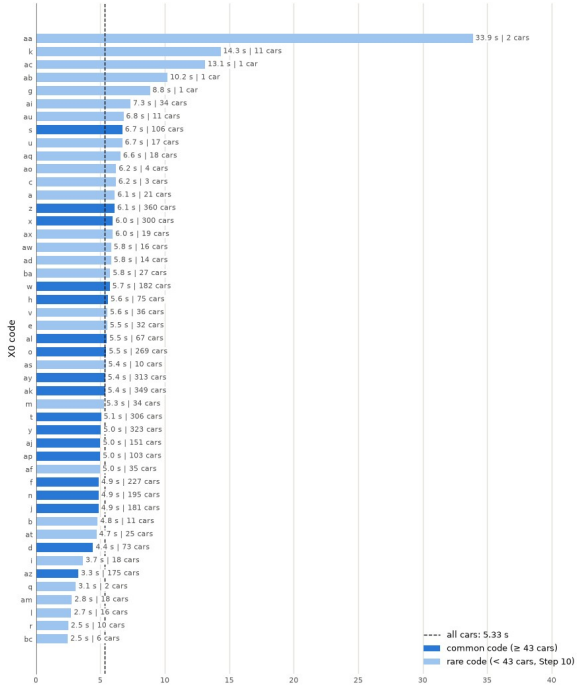
62 %

of cars: prediction at most 5 seconds off

91 %

of cars: at most 10 seconds off

- Tested on cars the model had not seen (5-fold cross-validation, 4,208 cars) (→ slide 33)
- The 10 biggest misses are all long tests (over 136 seconds), and all took longer than predicted (→ slide 35)
- Cars with rare X0 codes are not the problem: average miss 5.7 seconds vs 5.3 seconds for common codes



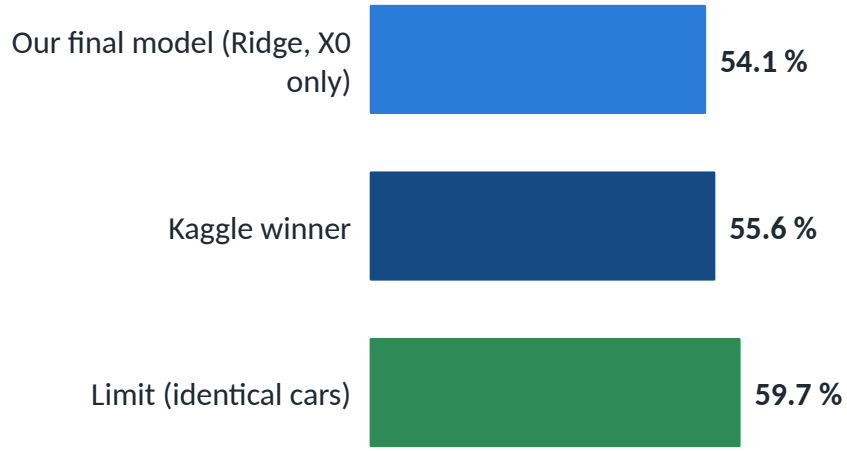
average miss per X0 code (seconds)

→ explained on slide 35 *Source: content list #20; out-of-fold test: Step 26 (each car predicted by the model that did not see it); 62.1 % / 90.8 %, 10 worst > 136.30 s: Steps 26–27 (R59–R63, D42); 5.72 s rare vs 5.29 s common: Step 27 (D42)*

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What is a good score?

Main · M3



share of the differences in test time explained, Kaggle's 4,209 hidden cars; limit explained on slide 23

≈ 97 %

of the winner's score (54.1 % / 55.6 %)

≈ 91 %

of the limit (54.1 % / 59.7 %; winner: ≈ 93 %)

In seconds (own test): typical miss 12.4 → 8.0 s, about 4.5 s better per car. Same average for every car = 0 %.

Source: content list #7; percentages + gain in seconds: changes 27, 28 (D64); winner 0.55550: D44 (verified on Kaggle by Mohammad; Gemini draft checked by Cowork); ours 0.54107: D47; limit 0.5965: Step 5; 93 % / 91 %: handover 23 Part B (D50)

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Data product: test-bench planner

Main · product

Car order

configuration (X0 code)

Prediction

Ridge with X0: about 4.5 s closer per car than the average (miss 12.4 → 8 s)

Bench plan

slot per car + about 10 s buffer (covers 91 of 100 cars)

Stakeholder (draft): test-bench planning engineer at the plant

Expert view (Mohammad Bajbouj, automotive R&D): plan each car with its predicted test time to reduce energy use in the end, and keep a buffer for extraordinary test times.

- The test itself does not get shorter – the plan gets more accurate (less waiting or overrun on the bench)
- Less energy and CO_2 = hypothesis (the data has no energy values) → pilot (→ slide 16)

Source: content list #22; stakeholder story = O6 (draft); 4.5 s: Steps 19–20; 91 % within 10 s: Step 26 (D42); expert view: his statement, change 29 (D64)

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Recommendations and next steps

Main · next

1

Plan with predictions

Expert view (M. Bajbouj): plan each car with its predicted time (about 4.5 s closer than the average) + about 10 s buffer (91 of 100 cars). Scale: about 300,000 cars a year ≈ 1,300 a day → about 1.6 h more accurate plan per day

2

Pilot (hypothesis)

One bench, one day, then one month: measure idle time and energy before and after planning with predictions

3

Record more (future work)

Long tests cannot be predicted from the configuration (→ slide 35) → record bench, tester and day; then test a targeted buffer in the pilot

4

Know the limits

Limit from 515 identical cars of 20 of 47 X0 types; the 57.5 % for X0 is measured on the same cars it describes; our own test and the Kaggle test use different cars

Source: content list #23; changes 30, 31 (D64); expert view: his statement; ~300,000 cars (Sindelfingen, 2017): Wikipedia 'Mercedes-Benz-Werk Sindelfingen'; 1,300/day = Cowork assumption 230 working days; 1.6 h = 4.5 s × 1,300; long tests: Step 29 (D61); ~8 s: Steps 20, 25; long tests: D42; 515 cars: Step 3 (D18); 20 of 47: handover 26 (D52, R72/D53); 57.5 % in-sample: Step 11 (R20); CV / Kaggle: D44

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M0 – the data: explanation slides

Backup for the M0 summary slide. Each slide names the main slide it belongs to and its source (content list #, notebook 04 step, decision log D-/R-IDs).

The problem

Many configurations

Almost every car is configured differently

Every car is tested

Each car runs on the test bench before delivery

Time = energy

Longer bench time means more energy and CO_2

Meaning: test times differ from car to car, which makes the test bench hard to plan.

→ *belongs to main slide 3* *Source: content list #2; problem statement, framing_h2_v3.md*

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Our question

Configuration

all 376 column values of one car



Predicted test time

seconds on the test bench

Can we predict each car's test-bench time from its configuration?

→ *belongs to main slide 3* *Source: content list #3; framing_h2_v3.md; hypothesis H2 chosen: D13*

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The data

Backup · M0

4,209
tested cars (train.csv)

376
anonymised columns: 8 text + 368 yes/no (0/1)

72–265 s
test time; shortest 72.11 s, longest 265.32 s

Meaning: Mercedes-Benz hid the column names (X0, X1 …) – we see the values, not what they mean.

→ *belongs to main slide 4* *Source: 4,209: Step 10 (R17); 376 = 8 + 368: Step 1 (R1, D17); 265 s: Step 9 (R13); 72.11 s, ID 3185: handover 23 Part B (D50)*

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M1 – EDA: explanation slides

Backup for the M1 summary slide. Each slide names the main slide it belongs to and its source (content list #, notebook 04 step, decision log D-/R-IDs).

How we found identical cars

1 Each car's 376 column values are written in one row → one configuration key per car
2 Cars with exactly the same key are identical (all 376 columns match) – a person can also check every group by hand

4,209
cars in total

3,694
one-of-a-kind cars

515
identical cars (12 %) in 217

cars per group	groups	cars
2 (pairs)	167	334
3	30	90
4	15	60
5	3	15
7	1	7
9	1	9
total	**217**	**515**

Largest group: 9 identical cars (X0 = x), test times 106–128 s – up to 22 seconds apart.
Second largest: 7 identical cars (X0 = n), test times 87.27–96.03 s – up to 8.8 seconds apart. Twins come from 20 of 47 X0 types.

→ *belongs to main slide 6* *Source: content list #5a; Steps 2–3 (R2, R3; D17, D18); 9-car group, 20 of 47: handover 26 (D52, D53); cars per size + 7-car group (IDs 689, 692, 735, 802, 821, 873, 964): handover 28 (D58); changes 12, 15 (D59)*

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From identical cars to the 60 % limit

1. Average test time of all 4,209 cars: 100.7 s. Spread around it: 160.8 s² (seconds squared, the variance – not an error). Its square root, 12.7 s, is the typical distance of a car from the average
2. Inside the 217 groups of identical cars the spread is still 64.9 s² (square root 8.1 s) – the configuration is the same, so this part is noise: reasons the data does not record, e.g. human error
3. Noise share: 64.9 s² out of 160.8 s² = 40 %. The other 60 % (95.9 s²) is the most the configuration can explain → limit about 60 %



total spread of all 4,209 cars: 160.8 s²

The 515 identical cars (12 %) are only the sample that measures the noise: they spread like all cars (12.6 s vs 12.7 s); with only these 515 cars the limit is 59 %. The noise (8.1 s) is close to our model's typical miss (about 8 s). Step by step: slide 24.

*→ **belongs to main slide 6** Source: content list #5b; 160.8 / 64.9 / 0.5965: Step 5 (R6, D18); 100.7 s, divisors, √ = 12.7 / 8.1 s: handover 28 (D58, R76 = all cars); 0.59: R6; 12.6 vs 12.7 s: handover 26; changes 13, 16 (D59)*

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How the noise is measured

1. For each group of identical cars: calculate the group's average test time
2. For each car: its distance to the group average (e.g. −4, 0, +4 s), then squared (16, 0, 16) – squaring stops + and − from cancelling
3. Add the squared distances of all 515 identical cars in all 217 groups: 19,332.3 s^2
4. Divide by 515 − 217 = 298: the 515 identical cars minus one value per group (217 groups), because each group's average is calculated from its own cars → 19,332.3 / 298 = 64.9 s^2
5. Compare with the spread of all 4,209 cars (160.8 s^2, slide 23): 64.9 / 160.8 = 40 % noise

Example (illustration only)

3 identical cars: 100 s, 104 s, 108 s

Average: (100 + 104 + 108) / 3 = 104 s

Distances: 100 − 104 = −4; 104 − 104 = 0; 108 − 104 = +4

Squared: $(-4)^2 + 0^2 + 4^2 = 16 + 0 + 16 = 32\ s^2$

Divide by 3 − 1 = 2: the 3 distances always add up to 0, so once 2 are known the 3rd is fixed → only 2 are free

32 / 2 = 16 s^2 → square root: 4 s

Equation (real data)
noise = sum of (test time − group average)² ÷ (identical cars − groups) = 19,332.3 ÷ (515 − 217) = 64.9 s^2

→ belongs to main slide 6 *Source: content list #5c; Step 5 (R4, R6; D18); 64.9 / 160.8 s^2: Step 5; 19,332.3 s^2, divisor 298: handover 28 (D58); example values made up for illustration; change 17 (D59)*

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What is X0?

- X0 is not an algorithm – it is one column of the data
- Mercedes-Benz hid the real column names, so they are called X0, X1, X2 …
- X0 is one of the 8 text columns: every car has exactly one X0 code; there are 47 codes
- For the models every X0 code becomes its own yes/no (0/1) column (one-hot encoding); a code never seen in training counts as an average car

Car	X0	X1	…	X10	X11	…	X385	test time
one car	bc	v	…	0	1	…	0	y (s)

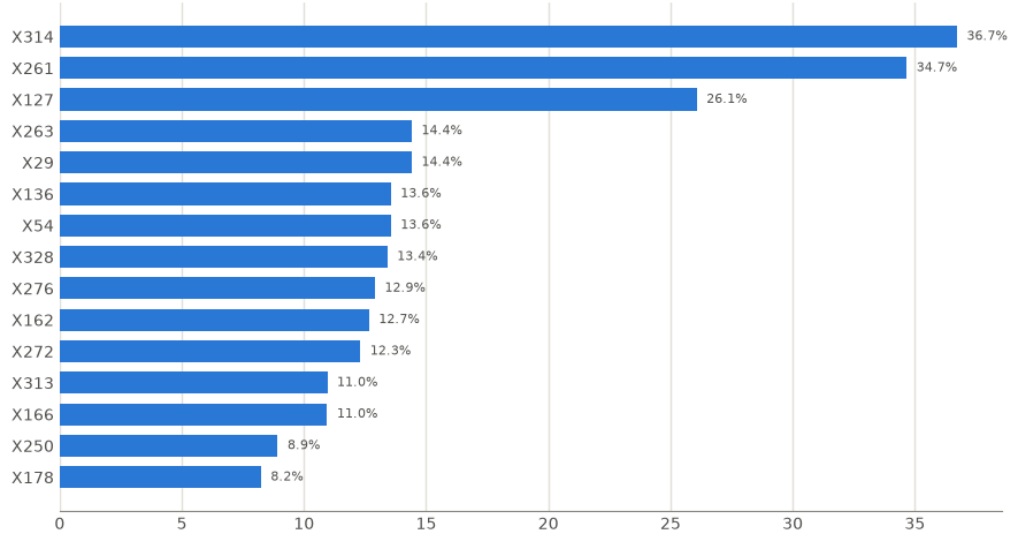
Example row – values for illustration only (text codes + yes/no (0/1) columns).

→ belongs to main slide 8 *Source: content list #9; 8 text columns: Step 1 (D17); 47 codes, e.g. bc / aa: Step 22 (D36); one-hot + unseen codes: Steps 12, 20 (R21, D25); unseen code → base value: Step 28 (R65, D46)*

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A strong column that tells nothing new

Backup · M1



X314	cars	avg time
with	1,817	109.5 s
without	2,392	94.0 s

X314 alone explains 36.7 %. But every X0 code is always with or always without X314 → once X0 is known it adds nothing (X0 alone explains 57.48 %; with X314 at most 57.52 %).

Lesson: a strong single link is not new information.

→ belongs to main slide 5 *Source: content list #11; X314: Step 15 (R29); 57.48 % → 57.52 %: Step 16 (R32, D29); bar chart: Step 15*

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X0 already decides X314

X0 code	X314 = 1	X314 = 0
z	0	360
ak	349	0
y	0	324
ay	313	0
t	0	306
x	300	0
o	0	269
f	0	227
n	0	195
w	182	0

Top 10 X0 codes by number of cars. The other 37 codes: also always one value.

0 of 47

X0 codes have both X314 values

- 14 X0 codes: X314 always 1 (1,817 cars)
- 33 X0 codes: X314 always 0 (2,392 cars)

Meaning: once X0 is known, X314 adds nothing (X0 alone 57.48 %, X0 + X314 57.48 %).

→ *belongs to main slide 5* *Source: content list #11a; table and counts: handover 27 result (D55, D56; R74); 57.48 % / +0.00 pp: Step 16 (D29)*

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Why we removed 0/1 columns

Backup · M1

368	– 12	– 45	– 8	303
yes/no (0/1) columns	never change	exact copies	exact opposites	kept

Outcome: the strong columns only repeat X0. All columns together did not beat X0 alone: share explained 57.4 % vs 58.7 %, typical miss 8.09 s vs 7.97 s (→ slide 30).

→ belongs to main slide 5 *Source: content list #12; 12 / 45 / 8 → 303: Steps 13–15 (R25, R28, R30; D27); repeat X0: Step 16 (D29); 0.574 / 8.09 s vs 0.587 / 7.97 s: Step 20 (R47, D32)*

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M2 – baseline: explanation slides

Backup for the M2 summary slide. Each slide names the main slide it belongs to and its source (content list #, notebook 04 step, decision log D-/R-IDs).

Feature engineering: only cleanup

Backup · M2

1. **Dropped the 265 s car**
   likely faulty test
2. **Left out X4**
   almost always the same value
3. **0/1 columns 368 → 303**
   constant, copies, opposites
4. **No new features, no scaling**
   inputs are 0/1 or codes

Where the 310 columns come from: 376 − X4 − 65 yes/no columns (12 never change, 45 copies, 8 opposites) = 7 text + 303 yes/no.

Why they made it slightly worse (8.09 s vs 7.97 s): they bring no new information beyond X0, but the model must learn hundreds of weights from 4,208 cars instead of 47 values → it learns some noise (overfitting). Ridge needed a strong penalty for them (alpha 10–100 vs 0.1).

Meaning: more columns did not help (57.4 % with all columns vs 58.7 % with X0 alone) → we keep X0 only.

→ belongs to main slide 9 *Source: content list #13; 310 + overfitting reason: change 24 (D64), alpha per fold: handover 28 (D58); 265 s car: Step 9 (D23); X4: Step 10 (R17, D25); 368 → 303: Steps 13–15 (D27); 0.574 vs 0.587: Step 20; X0 only: D40*

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M3 – model: explanation slides

Backup for the M3 summary slide. Each slide names the main slide it belongs to and its source (content list #, notebook 04 step, decision log D-/R-IDs).

Why Ridge, Random Forest and XGBoost



Ridge

one straight-line formula: each column adds a fixed number of seconds. It cannot see combinations (e.g. an X0 code that makes the test longer only together with a certain option)



Random Forest

many decision trees, averaged. Trees split the cars step by step, so they can catch combinations and uneven effects that Ridge misses



XGBoost

trees built one after another, each fixing the errors of the ones before – usually the strongest method for table data like ours

Meaning: if hidden combinations existed, Random Forest or XGBoost would score clearly better – they did not (58.9 % and 59.3 % vs 58.7 %): the limit is in the data, so the simplest model, Ridge, is enough (→ slide 12).

→ belongs to main slide 12 *Source: content list #17; requirement: D15; choice: D33; XGBoost install: D34; wording: change 34 (D66); 58.7 / 58.9 / 59.3 %: Steps 20, 23, 24*

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How the models were trained and tested

Backup · M3

Cross-validation is a testing method, not part of any model: the 4,208 training cars are split into 5 parts.

round 1	test	train	train	train	train
round 2	train	test	train	train	train
round 3	train	train	test	train	train
round 4	train	train	train	test	train
round 5	train	train	train	train	test

Training data: 4,208 cars (265 s car removed), 310 columns. Ridge: Step 20 · Random Forest: Step 23 · XGBoost: Step 24.

- Each round: train on 4 parts (about 3,367 cars), test on the 5th (about 842 cars) → every car is tested once by a model that did not see it
- The same 5 parts for Ridge, Random Forest and XGBoost → a fair comparison; settings chosen in a second cross-validation inside the 4 training parts
- Kaggle test: only the final Ridge (X0 only), retrained on all 4,208 cars, predicted the 4,209 hidden cars → explains 54.1 % (Kaggle holds their real times)

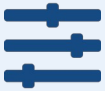
Meaning: all three models were trained and tested the same way; only the winner, Ridge, went to Kaggle.

→ belongs to main slide 12 *Source: Steps 17, 18 (KFold 5, shuffle, random_state 42), 20, 23, 24, 28: handover 28 (D58); final model D43; Kaggle 0.54107: D47; changes 19, 20 (D59)*

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Hyperparameter tuning

- Each model got a small grid of settings, tested inside the same 5-fold cross-validation – the test part is never used to choose settings
- The best settings were at the simple end of each grid
- A wider grid was not needed – no setting can pass the limit of about 0.60. Example Ridge: the penalty setting (alpha) 0.1 vs 1 changes the typical miss by only 0.007 s



Same 5 parts for every model and every setting

Ridge: alpha 0.1 / 1 / 10 / 100 → 0.1

Random Forest: depth 5 / 10 / no limit × leaf size 5 / 20 → depth 5

XGBoost: depth 2 / 4 × 200 / 500 trees → depth 2, 200 trees

→ belongs to main slide 12 *Source: content list #18; grids and chosen settings: Steps 20, 23, 24, handover 28 (D58); best at simple edge, no wider grid: R55–R56, D40; alpha 0.1 vs 1: Step 28 (R64, D46)*

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Can long tests be predicted?

0 of 9

identical twins of long-test cars were long too – they ran 26–58 s faster

- Same configuration, normal time → a long test comes from the test run itself (bench, tester, day), not from the configuration
- 12 X0 codes hold 43 of the 50 long tests – but also a third of all cars (1,395)
- So a warning rule is only about 2× better than chance

Warning rule (tested on unseen cars)	flagged	caught (of 49)	false alarms	right per flag
X0 code with above-average share	1,369	35	1,334	2.6 %
X0 code with ≥ 3× the share	578	17	561	2.9 %
Small model on all 310 columns	435	9	426	2.1 %
Chance (flag at random)	–	–	–	1.2 %

Long test = over 136.3 s (50 of 4,209 cars). Targeted buffer of 32.6 s: 14 of 49 long tests still get none, and more than 97 % of buffered cars would not need it.

Meaning: long tests cannot be predicted from the configuration → record test-run data and test a targeted buffer in a pilot (future work, slide 16).

→ *belongs to main slide 13* *Source: Step 29 (D61; R78–R81); handover 29 result (D60): Parts 1, 3, 4; future work: changes 31, 32 (D64)*

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